

Treatment Perceptions and Outcomes: Preliminary Results from Patient Audio and Self-Report Data in the PREDiCTOR Study

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BACKGROUND

- Self-report measures of therapeutic alliance provide quantitative indicators of patient-clinician relationships and have been linked to treatment disengagement as well as changes in patient symptoms and functioning.^{1, 2, 3} Unstructured patient speech about mental health care experiences offers an additional, qualitative source for understanding patient perceptions of providers.⁴
- The PREDiCTOR Study uses a multimodal approach to predict and subsequently prevent adverse outcomes in psychiatric care including treatment disengagement, emergency department visits, and hospitalization. Data types include a self-reported measure of therapeutic alliance, the Session Alliance Inventory (SAI),⁵ audio diary recordings, and semi-structured interviews with participants after six-months of participation.
- It is not yet known how self-reported therapeutic alliance and verbalized sentiments of care align and how these data modalities differentially relate to adverse events.

METHODS

Inclusion Criteria

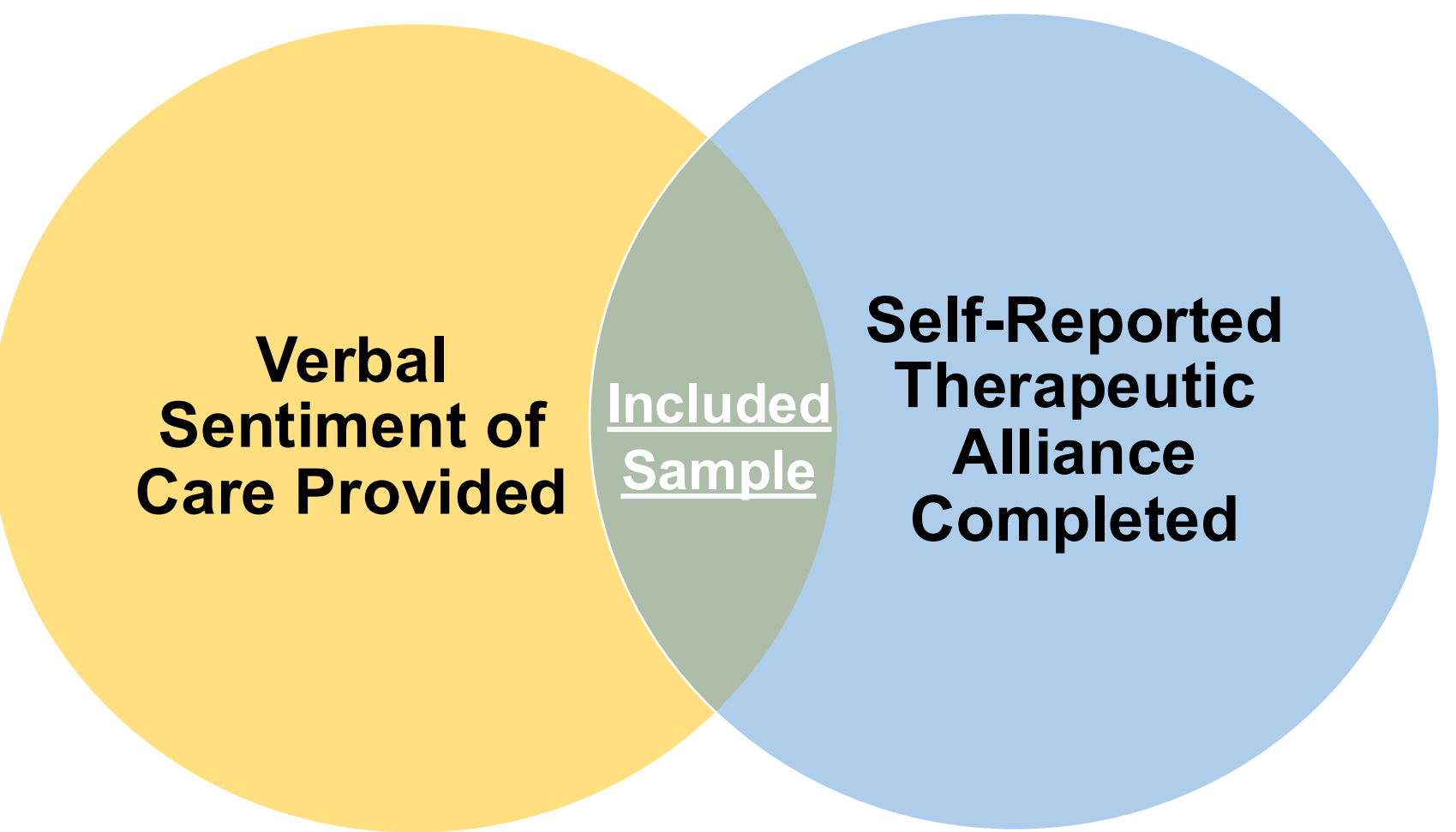


Figure 1. Inclusion Criteria. Participants with both data modalities completed but who did not initiate clinical care at Mount Sinai (no-showed to intake/screening or did not attend an appointment after intake/screening) were excluded (N = 5).

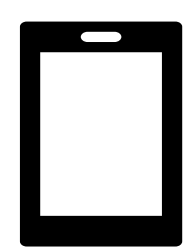
Demographics

Table 1. Demographic and clinical characteristics of the sample.

	N = 67		N = 67
Age, mean (SD)	30.75 (7.25)	Ethnicity	
Sex		Hispanic or Latino N (%)	20 (29.85%)
Male N (%)	24 (35.82%)	Non-Hispanic or Latino N (%)	47 (70.15%)
Female N (%)	43 (64.18%)		
Race		Primary Diagnosis	
Asian N (%)	8 (11.94%)	Depressive and Bipolar Disorders N (%)	33 (49.25%)
American Indian/Alaska Native N (%)	1 (1.49%)	Anxiety Disorders N (%)	17 (25.37%)
Black or African American N (%)	17 (25.37%)	Trauma-and Stressor-Related Disorders N (%)	8 (11.94%)
Native Hawaiian or Other Pacific Islander N (%)	1 (1.49%)	Substance Use Disorders N (%)	1 (1.49%)
White N (%)	30 (44.78%)	Schizophrenia Spectrum & other Psychotic Disorders N (%)	4 (5.97%)
Other N (%)	5 (7.46%)	Personality Disorders N (%)	1 (1.49%)
		ADHD and Externalizing Disorders N (%)	3 (4.48%)

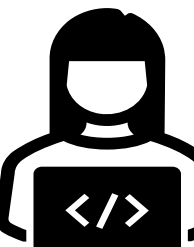
Data Modalities

Verbal Sentiment of Care



Audio Diaries

Participants provide unstructured speech through recording audio diaries in the MINDLamp2 mobile app. At times, participants expressed their feelings toward their mental health providers in these recordings. These audio diaries were identified for analysis. For participants who recorded multiple diaries about their care, the earliest recording was used.



6-Month Check-in Semi-structured Interviews

Participants also speak about their care during semi-structured interviews with research team members 6 months after providing informed consent. Sentiments were qualitatively coded as positive, neutral, or negative.

Table 2. Distribution of audio sources used as verbal sentiments for participants and number of participants within each sentiment group.

Verbal Source Used	Total	Positive Sentiment	Neutral Sentiment	Negative Sentiment
Audio Diary	38	11	19	8
6 Month Interview	29	19	8	2
Total	67	30	27	10

Adverse Events

Adverse events were operationalized as experiencing any of the following since the date of consent:

- Prematurely disengaging from psychiatric care
- Visiting the emergency department (ED) for a psychiatric concern
- Hospitalization for a psychiatric concern

Self-Reported Therapeutic Alliance

Table 3. Session Alliance Inventory (SAI) questions sent to PREDiCTOR participants after their psychiatry and therapy sessions at Mount Sinai. Participants rate each item using a 5-point Likert scale ranging from Strongly Disagree to Strongly Agree. SAI scores are out of 30 points, with two subdimensions, task/goal and bond, each of which are out of 15 points total.

Task/Goal Dimension	My clinician and I were working toward mutually agreed upon goals.
	We were in agreement about what is important for me to work on.
	I believe the way we were working with my problem(s) is correct.
Bond Dimension	I felt that my clinician cared about me even if I had done things that he/she does not approve of.
	I felt that my clinician appreciated me.
	My clinician and I respected each other.

For participants with more than one SAI completed, the SAI completed closest in time to the verbal sentiment of care was selected for the analysis. Preference was given to SAIs completed after the sentiment was expressed.

Table 4. Distribution of patient sample with adverse events since study enrollment. Three participants are counted in multiple groups as they experienced multiple types of adverse events.

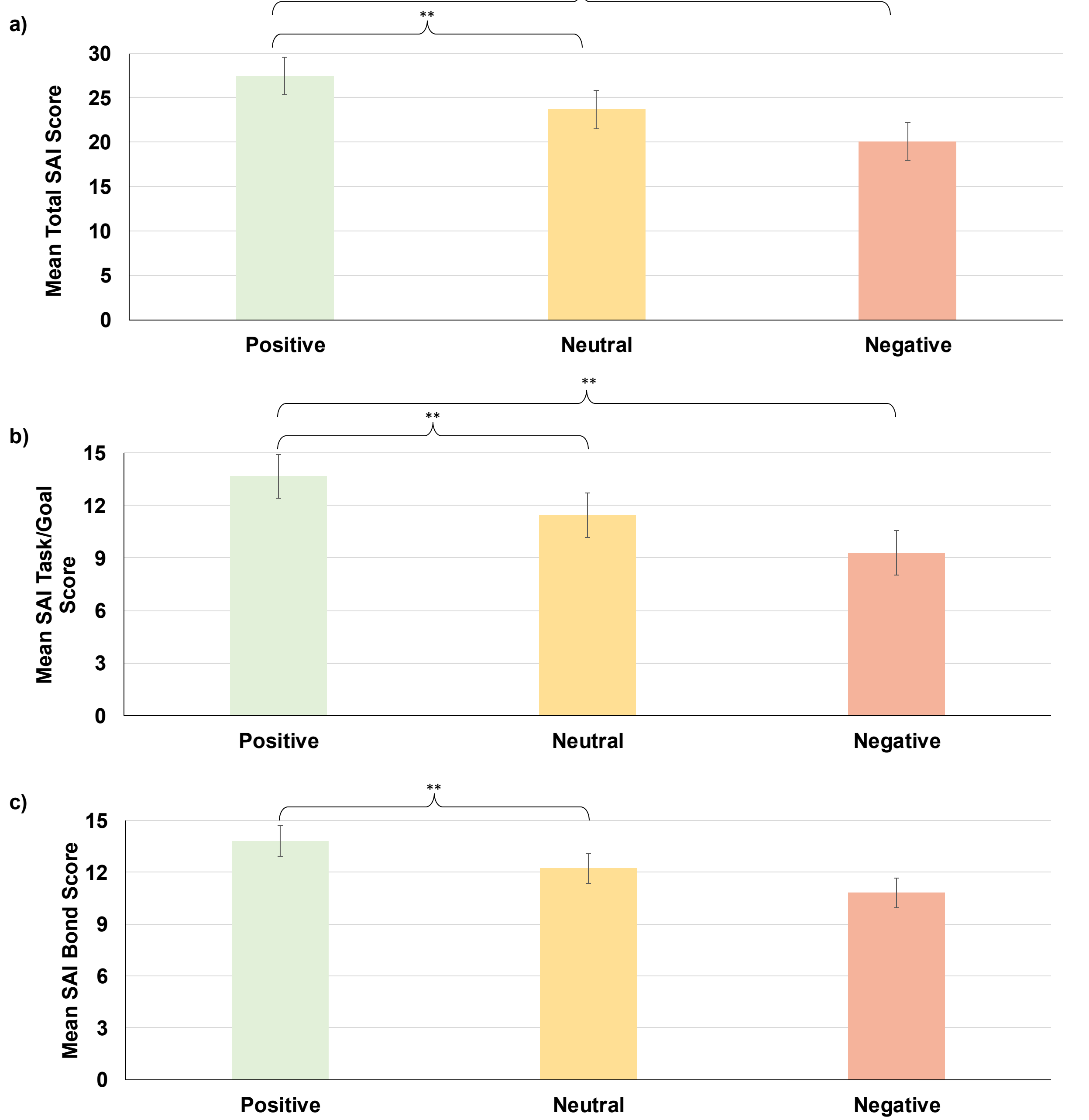
	Disengagement	ED Visit	Hospitalization	No Adverse Events
N (%)	8 (11.94%)	5 (7.46%)	4 (5.97%)	54 (80.60%)

AIMS

- Identify if self-reported therapeutic alliance scores as measured by the SAI differ significantly across groups of participants who reported positive, neutral, and negative verbal sentiments of care.
- Assess the relationship between SAI scores and the presence of patient adverse events.
- Assess the relationship between verbally expressed sentiments of care and adverse events.

RESULTS

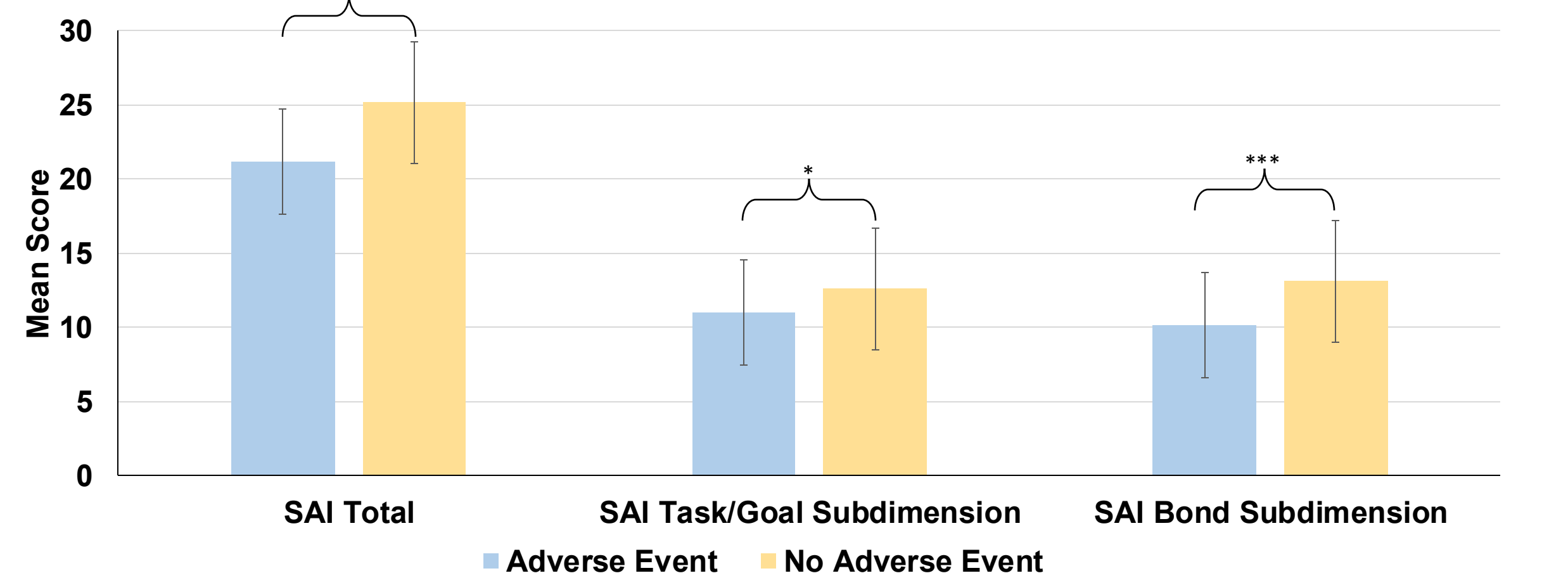
Relationship Between Self-Reported Therapeutic Alliance and Verbal Sentiments of Care



*p < 0.05 ; **p < .0.01 ; ***p < 0.001

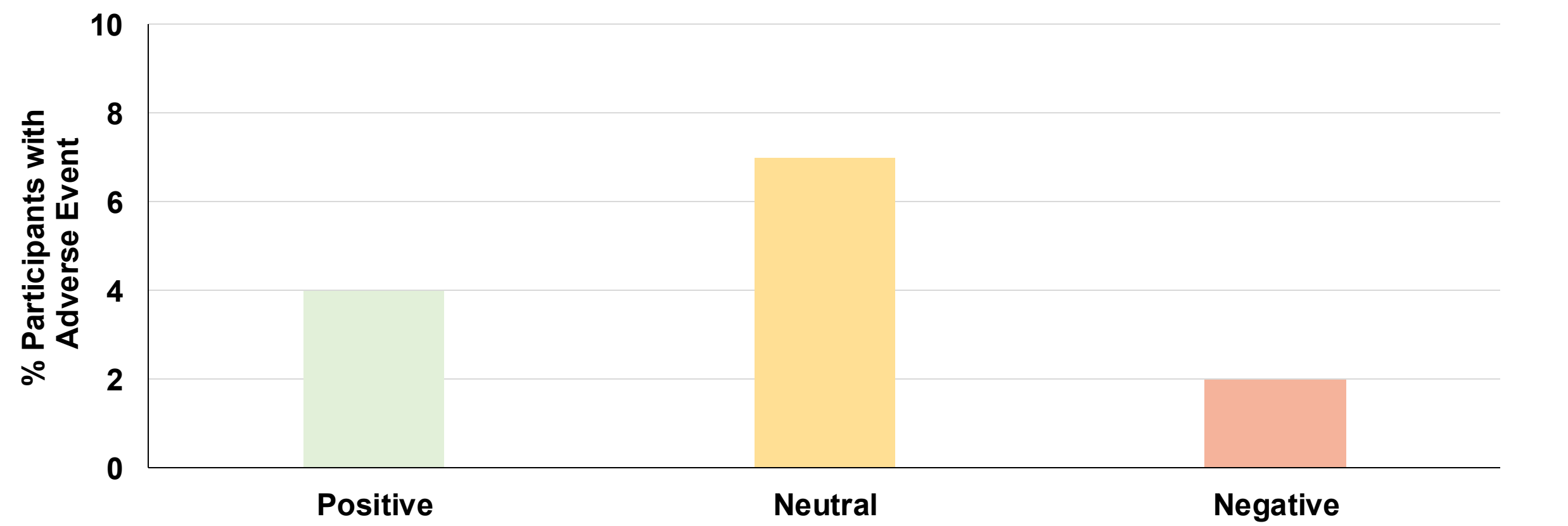
A non-parametric ANOVA revealed positive and neutral sentiment groups differed significantly across total SAI score ($W = -4.408$, $p = 0.0052$), task/goal subdimension ($W = -4.498$, $p = 0.0026$), and bond subdimension ($W = -3.637$, $p = 0.0274$). Positive and negative sentiment groups also differed significantly across total SAI score ($W = -4.276$, $p = 0.0071$) and task/goal subdimension ($W = -4.311$, $p = 0.0065$). An independent samples t-test showed a statistically significant difference in scores of participants who experienced adverse events and those who did not across total SAI ($T = 2.680$, $p = 0.0093$), task/goal subdimension ($T = 2.483$, $p = 0.0156$), and bond subdimension ($T = 2.558$, $p = 0.0129$). X^2 analysis showed no significant difference in percent of participants who experienced an adverse event across sentiment groups ($X^2 = 1.444$, $p = 0.4859$).

Self-Reported Therapeutic Alliance and Adverse Events



***p < 0.001

Sentiments of Care and Adverse Events



CONCLUSIONS AND FUTURE DIRECTIONS

- These findings suggest that verbally expressed sentiments toward care align with self-reported alliance measures. Additionally, findings show that self-reported alliance scores (total and across subdimensions) were significantly lower among patients who experienced adverse events, while no significant differences were observed for qualitative sentiments of care.
- Analyses will be repeated as the PREDiCTOR sample grows.
- We plan to develop models that use verbal sentiment and self-reported alliance measures to predict which patients will experience adverse outcomes over the course of care. Future work will examine changes in sentiment over time to provide indicators of treatment progress at the individual level and offer insight into how clinicians might intervene and modify approaches when necessary.

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References

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Acknowledgements

PREDiCTOR is funded by NIMH (1U01MH136535-01, 5U01MH136535-02).